AGENT-STATE: DURABLE CHECKPOINT LOG, VISIBLE BACKUP FAILURES, ONE BAD FILE NO LONGER BLOCKS AN ADAPTER

1 INTRODUCTION.

scripts/agent-state.sh copies adapter state from local disk ($AM_LOCAL) to the bucket (/data) every 15 s. From 2026-11-06 the claude adapter failed on every pass for a week; the only trace was a WARN on stdout, and a restart on 2026-11-13 restored a week-old tree. A failure is held to its bad paths, logged durably and shown in the app (figure 1).

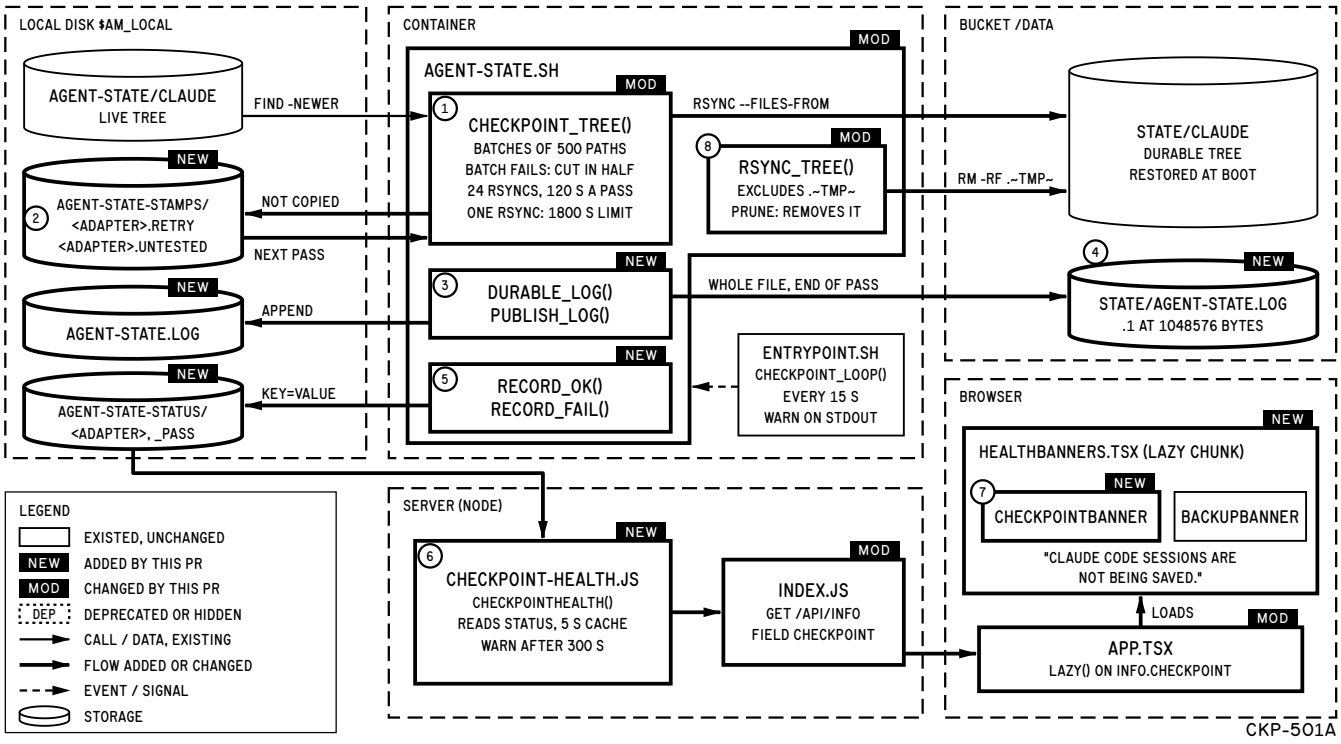


Figure 1. Agent-State Checkpoint, Layout and Data Flow

2 HOW IT WORKS.

Changed files go to rsync in batches of at most 500 paths. A batch that fails is cut in half, first half first, down to single paths; what copies is durable and the stamp moves on (1). Only a path that fails on its own stays in <adapter>.retry; paths a pass did not reach go to <adapter>.untested and into the next batches (2). durable_log() writes one line per failure (3); publish_log() copies the log to the bucket; at 1048576 bytes it rotates to .1 (4). record_ok() and record_fail() write each adapter's outcome (5). After 300 s without a clean pass checkpointHealth() lists the adapter in /api/info as checkpoint (6), and the app shows a line that cannot be dismissed (7). .~tmp~ leftovers are excluded and pruned (8) (figure 2).

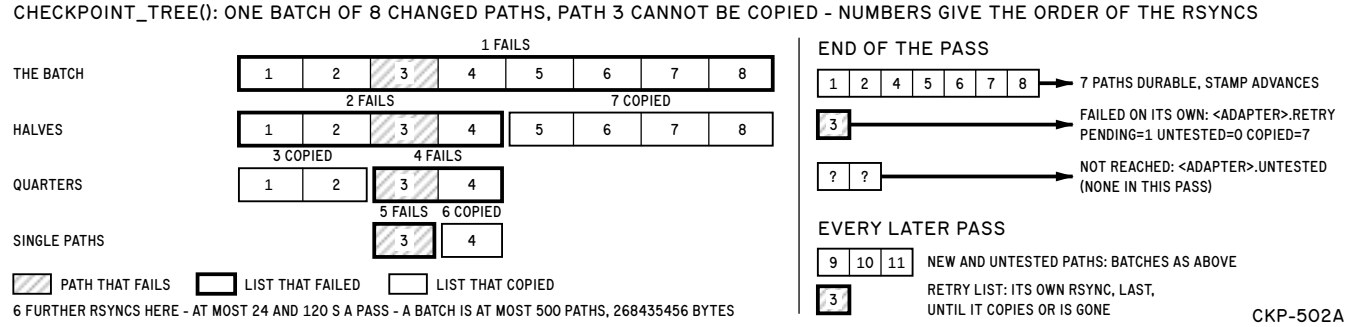


Figure 2. Halving of a Failed List in checkpoint_tree()

3 DATA.

Table 1. Checkpoint Limits and Defaults

Parameter	Value
AGENT_STATE_BATCH_FILES (paths in one rsync)	500
AGENT_STATE_BATCH_BYTES (bytes in one rsync)	268435456 bytes
AGENT_STATE_RETRY_MAX (further rsyncs in a pass)	24
AGENT_STATE_RETRY_SECONDS (halving, in a pass)	120 s
AGENT_STATE_CHECKPOINT_TIMEOUT (one rsync)	1800 s
AGENT_STATE_RETRY_SLOW_SECONDS (retry after a timeout)	3600 s
AGENT_STATE_LOG_MAX (log rotates to .1)	1048576 bytes
AGENT_STATE_LOG_EVERY (unchanged failure logged again)	20 passes
AM_CHECKPOINT_WARN_SECONDS (no clean pass, then warn)	300 s

CAUTION

Why the passes failed from 2026-11-06 is not known. This PR makes the next failure logged and visible; it does not prove the old cause fixed.

VERIFY CHECKPOINTS — **OPERATOR**

```
1 LOG - read the durable log
  √ /data/state/agent-state.log
2 INFO - curl -s localhost:$AM_PORT
    /api/info | jq .checkpoint
  √ null - every adapter is saved
3 DIR - $AM_LOCAL/agent-state-status
  √ failures=0 in each file
4 APP - no warning line shown
```

CKP